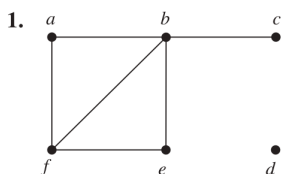


Mat 2540 HW 11

let $|V| = \# \text{ of vertices}$

$|E| = \# \text{ of edges}$

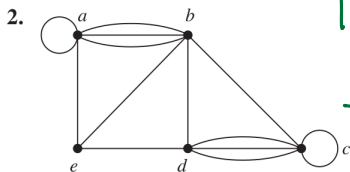
In Exercises 1–3 find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.



$|V| = 6$ $|E| = 6$, d is isolated
c is pendant

	a	b	c	d	e	f
deg	2	3	1	0	2	3

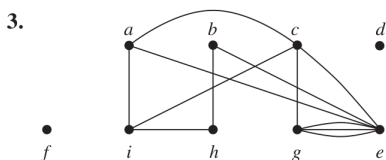
$\Rightarrow$ sum of degree = 12 and
it equals $2|E|$



$|V| = 5$, $|E| = 13$ No isolated or
pendant vertex

	a	b	c	d	e
deg	6	6	6	5	3

$\Rightarrow$ sum of degree = 26 and
it equals $2|E|$



$|V| = 9$, $|E| = 12$, f, d is isolated

	a	b	c	d	e	f	g	h	i
deg	3	2	4	0	6	0	4	2	3

$\Rightarrow$ sum of degree = 24 and
it equals $2|E|$

5. Can a simple graph exist with 15 vertices each of degree five?

If there were a graph with 15 vertices each of degree 5, then
the sum of degree is $15 \cdot 5 = 75$.

Since the half of sum of degree equals the number of edges,
it implies there were $\frac{75}{2} = 37.5$ edges which is impossible.

Therefore, this graph doesn't exist.

6. Show that the sum, over the set of people at a party, of the number of people a person has shaken hands with, is even. Assume that no one shakes his or her own hand.

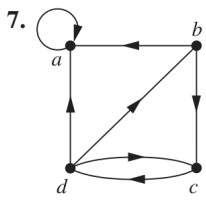
We can build an undirected graph and put an edge between two people
if they shake hand with each other and "the degree of a person"
represents the number of people this certain person has shaken
hands with.

Since the sum of the degrees is an even, then the sum
of shaking hands is even as well.

In Exercises 7–9 determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

10. For each of the graphs in Exercises 7–9 determine the sum of the in-degrees of the vertices and the sum of the out-degrees of the vertices directly. Show that they are both equal to the number of edges in the graph.

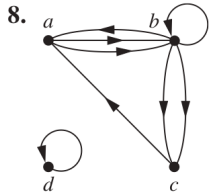
Let $|V| = \# \text{ of vertices}$, $|E| = \# \text{ of edges}$



$$|V| = 4, |E| = 7$$

	a	b	c	d
in-deg	3	1	2	1
out-deg	1	2	1	3

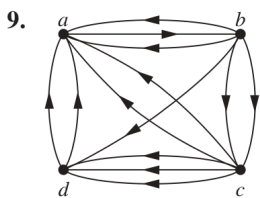
sum of in-degree = 7 both equal $|E|$
 $\Rightarrow$ sum of out-degree = 7



$$|V| = 4, |E| = 8$$

	a	b	c	d
in-deg	2	3	2	1
out-deg	2	4	1	1

sum of in-degree = 8
 $\Rightarrow$ sum of out-degree = 8
 and both equal $|E|$



$$|V| = 4, |E| = 13$$

	a	b	c	d
in-deg	6	1	2	4
out-deg	1	5	5	2

sum of in-degree = 13
 $\Rightarrow$ sum of out-degree = 13
 and both equal $|E| = 13$

20. Draw these graphs.

a) K_7

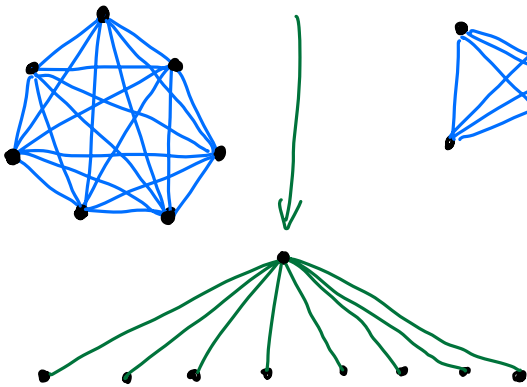
b) $K_{1,8}$

c) $K_{4,4}$

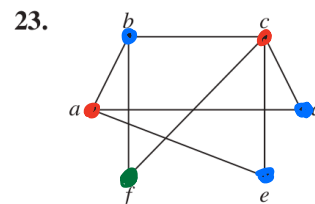
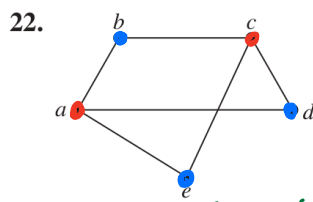
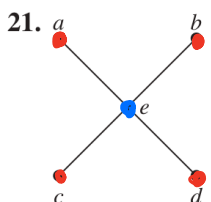
d) C_7

e) W_7

f) Q_4



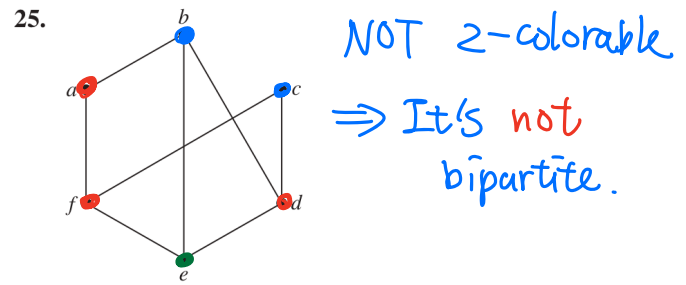
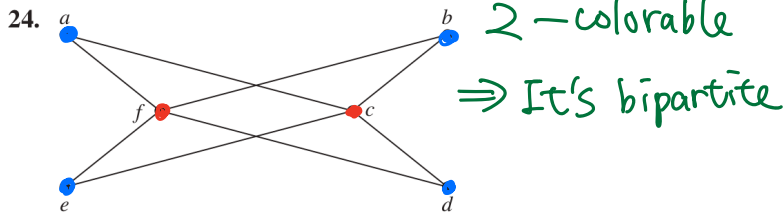
In Exercises 21–25 determine whether the graph is bipartite. You may find it useful to apply Theorem 4 and answer the question by determining whether it is possible to assign either red or blue to each vertex so that no two adjacent vertices are assigned the same color.



Not 2-colorable
 $\Rightarrow$ It's not bipartite.

2-colorable
 $\Rightarrow$ It's bipartite

2-colorable
 $\Rightarrow$ It's bipartite



37. How many vertices and how many edges do these graphs have?

a) K_n

b) C_n

c) W_n

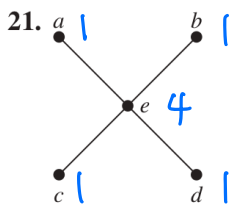
d) $K_{m,n}$

e) Q_n

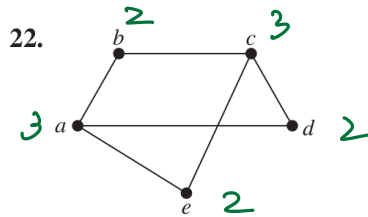
- a) K_n has n vertices and $\frac{n(n-1)}{2}$ edges
 b) C_n has n vertices and n edges
 c) W_n has $n+1$ vertices and $2n$ edges
 d) $K_{m,n}$ has $m+n$ vertices and $m \cdot n$ edges
 e) Q_n has 2^n vertices and 2^n edges

The **degree sequence** of a graph is the sequence of the degrees of the vertices of the graph in nonincreasing order. For example, the degree sequence of the graph G in Example 1 is 4, 4, 4, 3, 2, 1, 0.

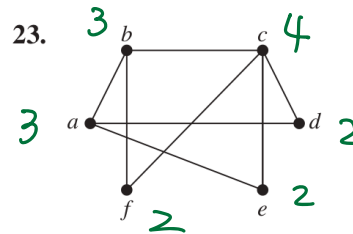
38. Find the degree sequences for each of the graphs in Exercises 21–25.



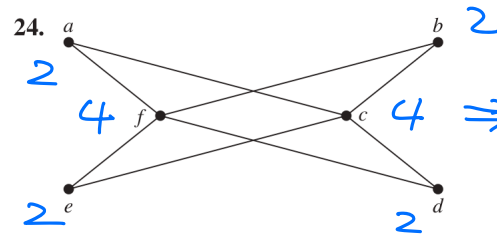
$\Rightarrow 4, 1, 1, 1, 1, 1$



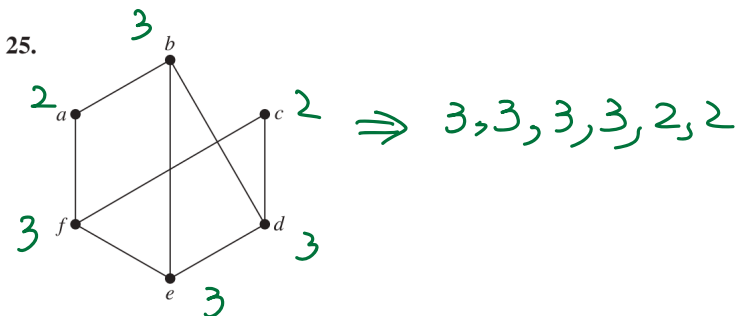
$\Rightarrow 3, 3, 2, 2, 2, 2$



$\Rightarrow 4, 3, 3, 2, 2, 2$



$\Rightarrow 4, 4, 2, 2, 2, 2$



$\Rightarrow 3, 3, 3, 3, 2, 2$

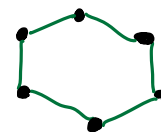
A sequence $d_1, d_2, \dots, d_n$ is called **graphic** if it is the degree sequence of a simple graph.

44. Determine whether each of these sequences is graphic. For those that are, draw a graph having the given degree sequence.

- a) 5, 4, 3, 2, 1, 0 b) 6, 5, 4, 3, 2, 1 c) 2, 2, 2, 2, 2, 2

b) NOT Graphic. There are six vertices and it is impossible for a vertex with degree 6 (which has 6 edges)

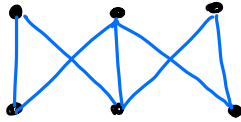
a) NOT Graphic. There are six vertices and it is impossible for a vertex with 5 edges when there is an isolated vertex (0 degree) in a simple graph
 c) Graphic and it is a C_6



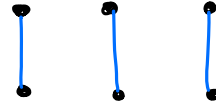
d) 3, 3, 3, 2, 2, 2 e) 3, 3, 2, 2, 2, 2 f) 1, 1, 1, 1, 1, 1

d) NOT Graphic, sum of degree is an odd number which is impossible

e) Graphic

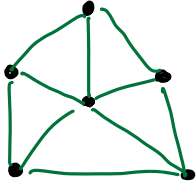


f) Graphic



g) 5, 3, 3, 3, 3, 3 h) 5, 5, 4, 3, 2, 1

g)



Graphic which is a W_5

h) Not Graphic.

There are six vertices and two vertices are have 5 edges but it is impossible to have one vertex with only one edge.

61. The **complementary graph** $\overline{G}$ of a simple graph G has the same vertices as G . Two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G . Describe each of these graphs.

a) $\overline{K_n}$

b) $\overline{K_{m,n}}$

c) $\overline{C_n}$

d) $\overline{Q_n}$

a) K_n is a complete graph and there is no way to add more edge(s).
Thus, $\overline{K_n} = \emptyset$

b) $K_{m,n}$ is bipartite with m vertices have edges with each vertex from all the other n vertices. Then $\overline{K_{m,n}}$ is a graph combining with a K_m and K_n .

c) $\overline{C_n} = K_n - C_n$